Infosys Springboard Virtual Internship 7.0
Completion Report

Team Details : Nandini Verma, Sasi Madhuri, Rahul Yadha, M Madhumitha, Nakka Rajeev

Batch Number 01 : Enterprise Learning Platform
Start date : 29-06-2026
Name: Nandini Verma
Internship Duration: 40-day (12 Weeks)

1. Project Title

Enterprise Learning Platform

2. Project Objective

Enterprise Learning Platform is an AI-powered enterprise learning platform designed to help users monitor and improve their technical skills, career readiness, productivity, and daily learning habits. The main objectives of this project are:

• To provide users with an interactive AI chatbot and code tutor for learning support.
• To analyze user learning data and provide insights into their skill patterns.
• To predict mood, stress levels, productivity, and career elevation paths using machine learning & AI models.
• To provide personalized learning suggestions to improve technical and career performance.
• To create a dashboard that visualizes user learning analytics in an easy-to-understand format.

This project aims to combine artificial intelligence and data analytics to promote better skill awareness and learning management.

3. Project description in detail

Enterprise Learning Platform is a web-based AI enterprise learning platform developed using React 19, Spring Boot 3.5 (Java 17), MySQL 8.0, and Google Gemini AI. The platform allows users to track their daily learning activities, analyze skill progress data, and receive intelligent recommendations.

The system consists of multiple modules that work together to provide a complete learning monitoring experience.
Landing Page:

The landing page introduces users to the platform and highlights the key features of Enterprise Learning Platform. It provides information about the purpose of the platform and encourages users to register or log in.

Authentication System:

The system includes a secure login and registration mechanism using JWT sessions and Google OAuth2 SSO that allows users to access their personal learning dashboard.

Learning Dashboard:

The dashboard is the main interface where users can view their learning insights. It displays charts and statistics about course progress, XP points, study streaks, and skill badges.

AI Chatbot:

The chatbot interacts with users and provides technical advice based on their inputs. It acts as a virtual assistant and code tutor that can guide users in maintaining a disciplined learning path.

Machine Learning Model & AI Career Engine:

The platform uses machine learning and AI models built using Scikit-learn and Google Gemini to predict skill growth, stress levels, and productivity. The model is trained on a learning dataset that includes features such as: • Course completion rate

• Quiz evaluation scores

• Coding sandbox submission accuracy

• Daily study streak duration

• Skill gap benchmark score

The predictions help users understand their technical competency and career path readiness.

Admin Panel & Instructor Studio:

An admin panel allows administrators to monitor system activity, manage users, review learning analytics, and control platform access. The entire system is designed to provide a simple, user-friendly interface while using advanced technology in the backend.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project objective is decided, making github repository	Github repository created, finalizing teams with the project overview
Week 2	Tech stack evaluation, architecture planning	Spring Boot & React setup completed, learning about REST APIs
Week 3	Setup user authentication, JWT tokens	Done with verified signin | login with JWT sessions & OAuth2
Week 4	Learning about API's, setting api's for project, database design	API's is setup using Gemini AI, OpenAI etc. which makes project effective
Week 5	LMS pipeline development – course hierarchy, content mapping	Tiered course catalog (Basecamp, Ridge, Summit) implemented
Week 6	Spring Boot service logic, repository interfaces	Implementing service layer functions and MySQL JPA mappings
Week 7	Gemini AI integration, prompt template design	Used Gemini models for live AI assistant & code tutor in Enterprise Learning Platform
Week 8	Quiz assessment, assignment portal development	Automated quiz scoring and assignment upload portal completed
Week 9	Gamification system - XP, badges, leaderboards	Displayed leaderboards, XP points, and user badges based on activity

Week	Activities Planned	Activities Completed
Week 10	Dark/Light theme switch, UI polishing	Theme toggle working across UI, modern glassmorphism aesthetic
Week 11	Dashboard design (like user activity, streak, profile, feedback)	Implementing dashboard with different course tiers and activity heatmap
Week 12	Adding career suite (ATS Resume, digital certs, admin panel)	Project fully tested and ready for evaluation

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Defined the project objectives, scope, system architecture, and responsibilities for developing SkillSphere platform.	29 Jun 2026
Prototype / First Draft	Developed initial working prototype including Spring Boot backend, basic UI pages, and core course input functionality.	08 Jul 2026
Mid-Term Review	Evaluated project progress, improved dashboard design, integrated Gemini AI model, and refined LMS functionality.	23 Jul 2026
Final Submission	Completed the fully functional SkillSphere system including dashboard analytics, AI chatbot, gamification, and certs.	13 Aug 2026
Final Presentation	Demonstrated the working system highlighting features such as skill tracking, AI roadmaps, chatbot, and architecture.	01 or 02 Sep 2026

5b. Project execution details

1. Requirement Analysis:

The project started with identifying the main objectives of the Enterprise Learning Platform system, which included building an AI-powered learning assistant that can track user progress data, provide chatbot support, and generate predictions related to skills, streak, and career readiness.

2. System Architecture Design:

The architecture of the system was planned using a client-server model where the React SPA frontend interacts with the Spring Boot backend REST controllers. The backend processes user requests, communicates with the MySQL database, and integrates the Google Gemini AI engine.

3. Frontend Development:

The user interface was developed using HTML, CSS, JavaScript, and React 19. Multiple pages were created including the landing page, login page, dashboard page, course player, and chatbot interface to provide a smooth and user-friendly experience.

4. Backend Development:

The backend was implemented using Spring Boot 3.5 and Java 17. It handles user authentication, manages API routes, processes data, connects to the MySQL database, and communicates with the AI module.

5. Database Integration:

MySQL database was used to store user credentials, course records, quiz scores, and prediction data. This ensures that all user information and analytics can be accessed and managed efficiently.

6. Machine Learning & AI Integration:

Google Gemini AI and Scikit-learn models were integrated into the Spring Boot backend so that AI tutoring responses and career roadmaps can be generated in real-time when users input their learning data.

7. LMS & Interactive Playground Integration:

Structured Basecamp, Ridge, and Summit course routes were built with video streaming, markdown reading, and interactive coding exercise sandboxes.

8. Dashboard Development:

An interactive dashboard was created using Chart.js to visualize learning data through graphs and charts. This helps users easily understand their progress trends and performance.

9. Chatbot Integration:

An AI chatbot module was added to allow users to interact with the system. The chatbot provides technical advice and guidance based on user inputs and evaluation results.

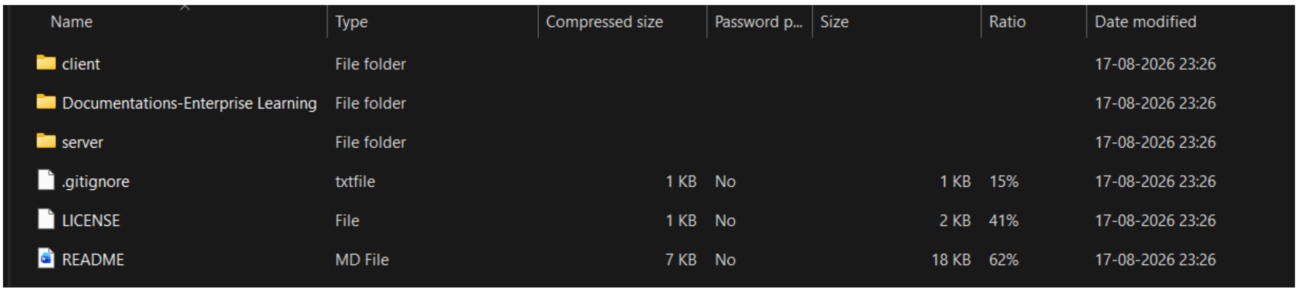
10. Admin Panel and Testing:

An admin panel was implemented to monitor users and manage system activities. Finally, the entire system was tested to ensure that all modules work correctly and integrate smoothly.

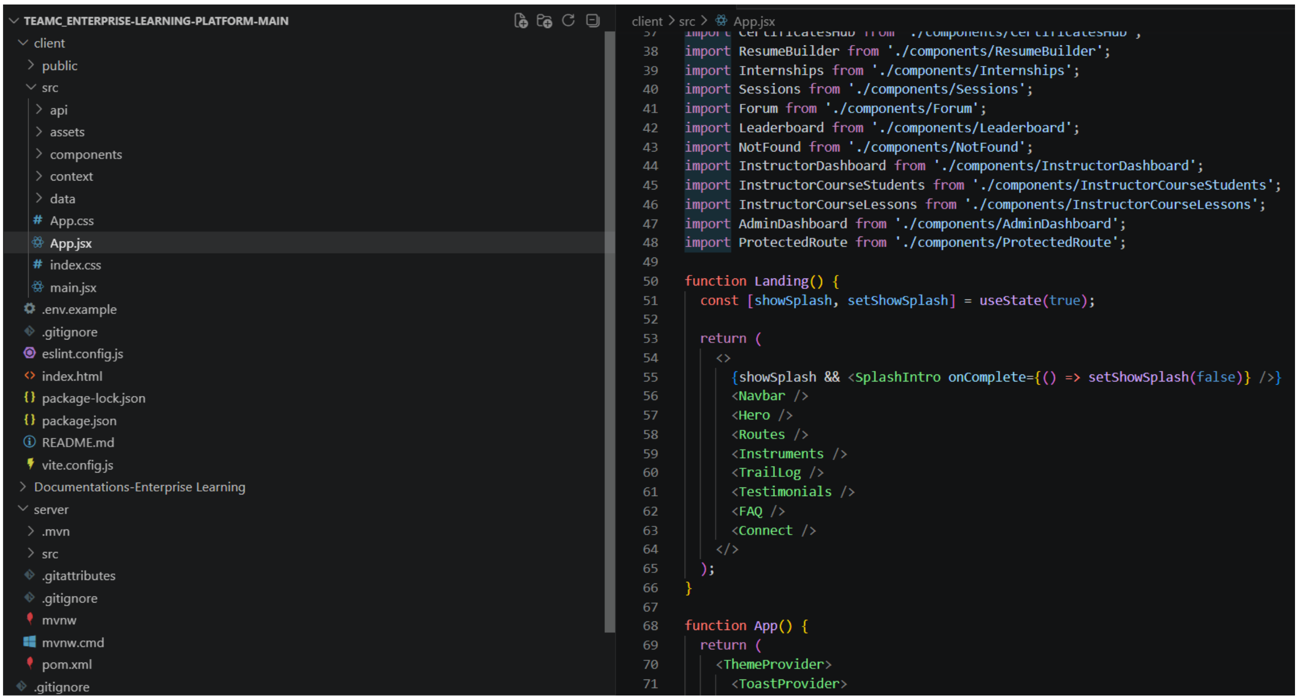
6. Snapshots / Screenshots

Opening the project folder containing all the main files such as ServerApplication.java, SecurityConfig.java, controllers, services, App.jsx, and components in VS Code.

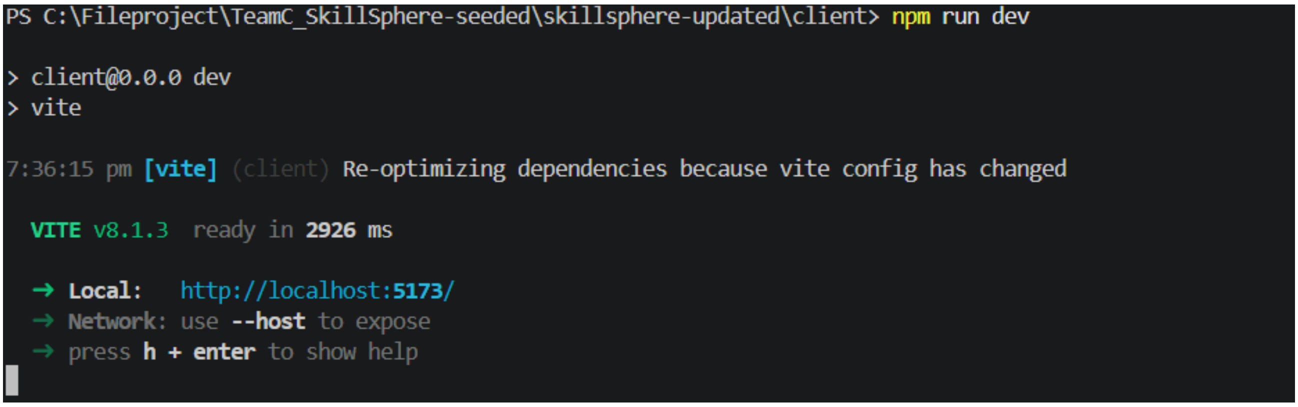
1. Opening the main backend and frontend application source files in VS Code to view the full-stack code.

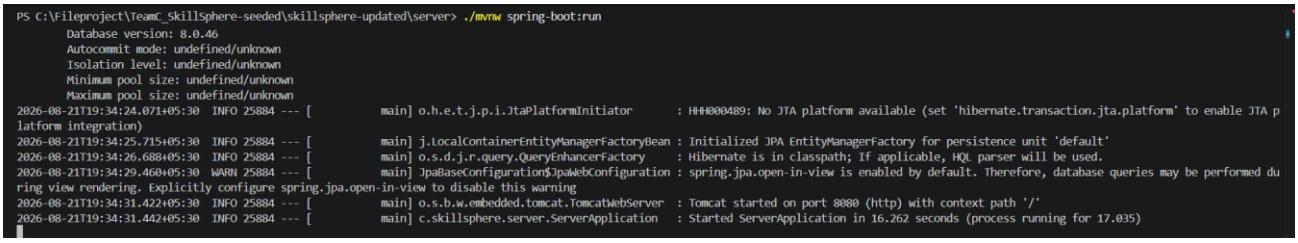


Name	Type	Compressed size	Password p...	Size	Ratio	Date modified
client	File folder					17-08-2026 23:26
Documentations-Enterprise Learning	File folder					17-08-2026 23:26
server	File folder					17-08-2026 23:26
.gitignore	txtfile	1 KB	No	1 KB	15%	17-08-2026 23:26
LICENSE	File	1 KB	No	2 KB	41%	17-08-2026 23:26
README	MD File	7 KB	No	18 KB	62%	17-08-2026 23:26



1. Running the application in the terminal of VS code (./mvnw spring-boot:run and npm run dev).





2. Copying or clicking the local server link provided by Vite (http://localhost:5173 / http://localhost:8080). The home page of the Enterprise Learning Platform appears on the browser.

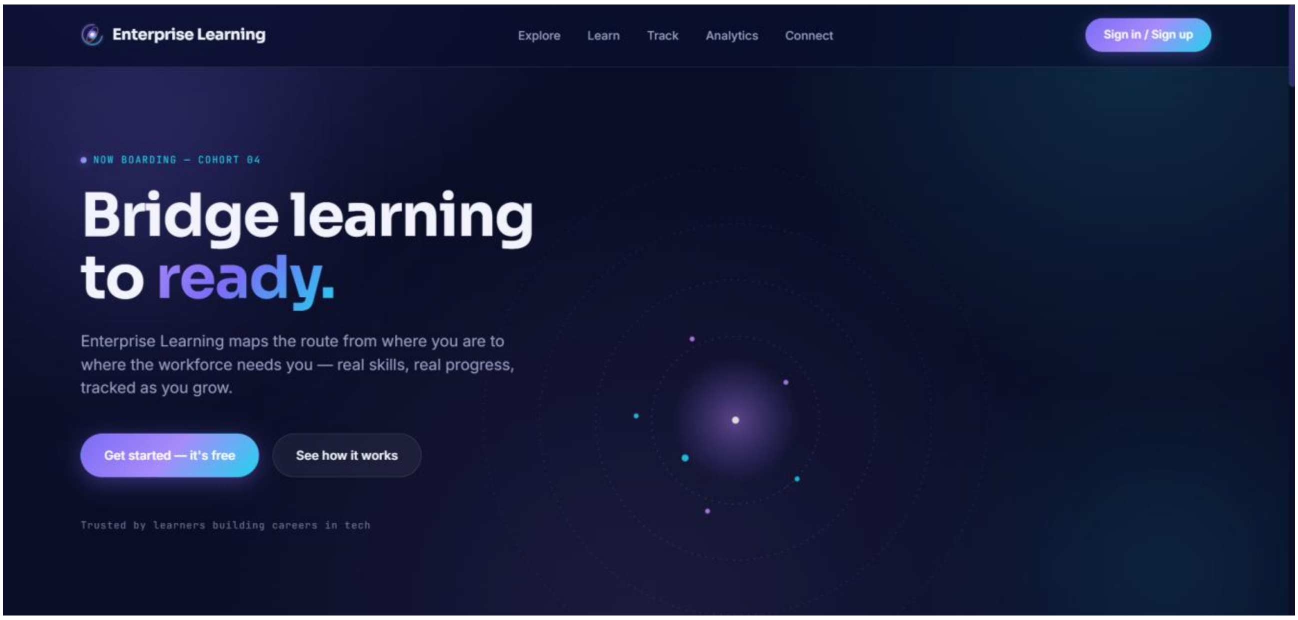
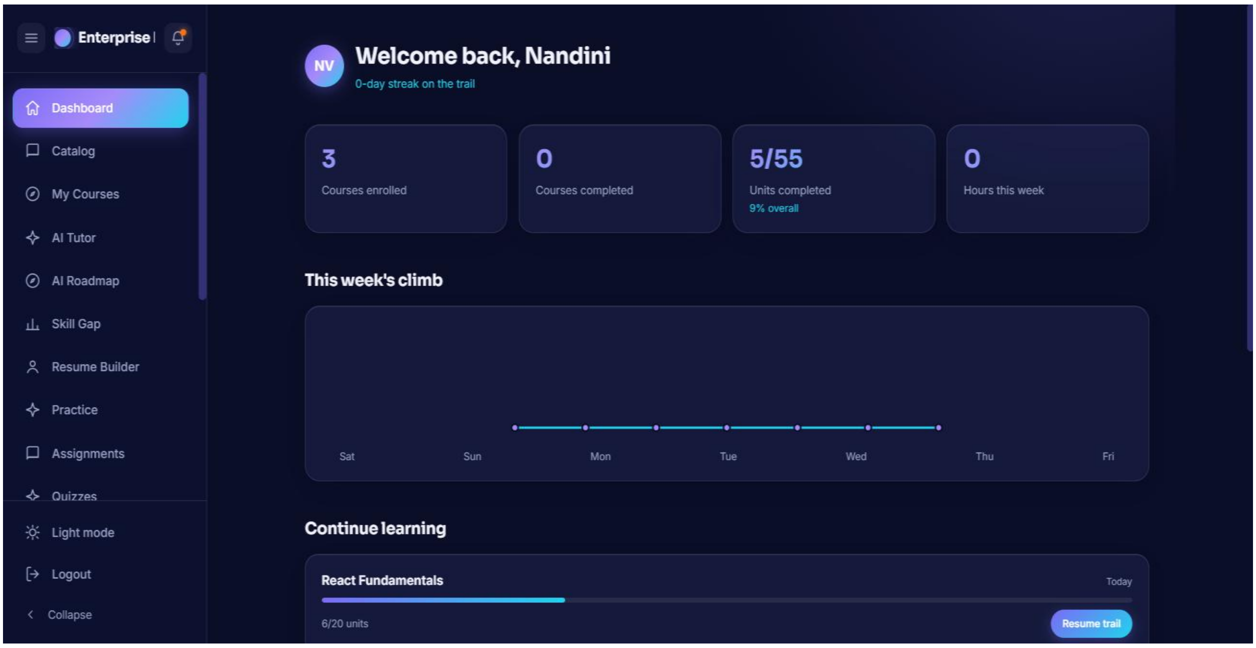
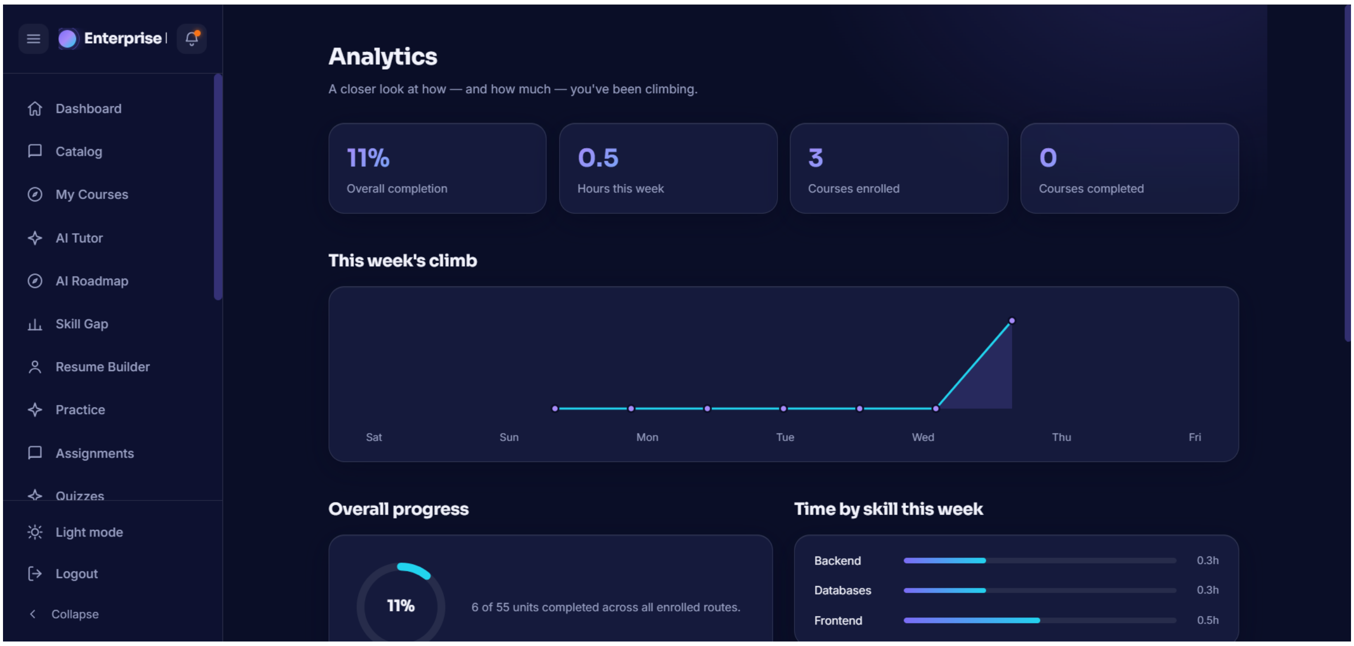


Figure 1: Landing Page Interface

LANDING PAGE

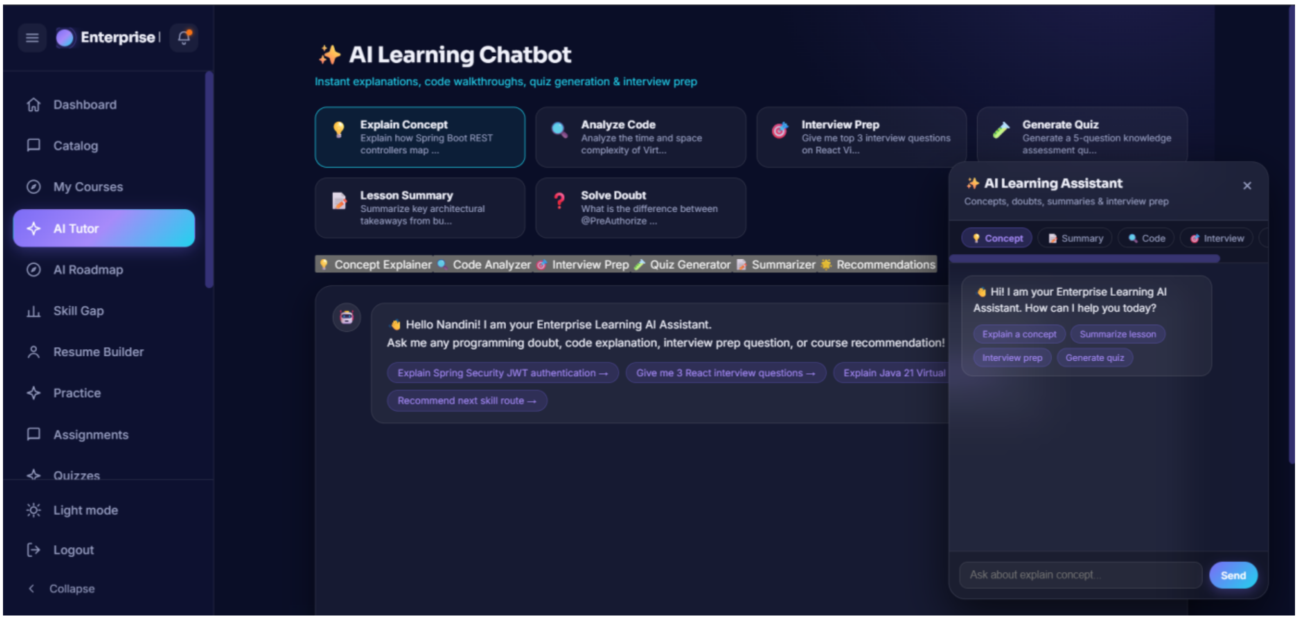
3. Next page includes log in and also create an account for the new user or you can start your account with google also.





Student Analytics & Learning Dashboard Interface

4. Chatting with AI tutors and code assistants is also available in the application.



AI Assistant & Code Complexity Analysis Interface

7. Challenges Faced

During development of the enterprise learning platform, several technical and operational challenges were encountered:

1. Integrating AI Models with the Web Application:

One of the main challenges was connecting the Google Gemini AI engine with the Spring Boot backend so that tutoring predictions and responses could be generated dynamically based on user inputs.

2. Designing a User-Friendly Dashboard:

Creating an interactive and visually clear dashboard that displays learning analytics in an understandable format required careful UI design and data visualization planning.

3. Handling Data Processing and Predictions:

Preparing the dataset and ensuring accurate recommendations for student roadmaps using AI models was a complex task.

4. Database Management:

Managing user data securely and maintaining proper database connectivity with MySQL while storing course records required careful implementation.

5. Chatbot Integration:

Integrating the chatbot system and ensuring it responds correctly to technical queries and code syntax was another challenge.

6. Frontend and Backend Communication:

Ensuring smooth communication between the frontend React interface and Spring Boot backend through REST APIs required debugging and testing.

7. System Testing and Debugging:

During development, several issues such as server errors, data handling problems, and interface bugs were encountered and needed to be resolved through testing.

8. Ensuring Smooth User Experience:

Making sure that all modules such as login, dashboard, chatbot, and predictions worked seamlessly together was a significant challenge during the final integration phase.

8. Learnings & Skills Acquired

1. Java & Spring Boot Programming:

Understanding Java 17 and Spring Boot 3.5 was essential for developing the backend logic of the application and implementing REST services.

2. Spring Security & JWT Framework:

Knowledge of Spring Security helped in building the web application backend, managing routes, handling JWT authentication, and integrating OAuth2.

3. Frontend Development:

Skills in HTML, CSS, JavaScript, and React 19 were required to design responsive and user-friendly interfaces such as the landing page, login page, dashboard, and chatbot page.

4. Database Management:

Knowledge of MySQL 8.0 and Spring Data JPA was required to store and manage user information, course records, and system data efficiently.

5. Machine Learning & AI Concepts:

Understanding AI concepts such as prompt engineering, REST API integration, and fallback mechanisms was necessary to build the AI code tutor.

6. Google Gemini API:

The Google Gemini 1.5 Flash API was used to develop and train intelligent response workflows for technical concepts.

7. Data Visualization:

Knowledge of visualization tools like Chart.js helped in displaying learning analytics through charts and graphs on the dashboard.

8. Problem Solving and Debugging:

Strong debugging and problem-solving skills were needed to fix errors, integrate different components, and ensure smooth functionality of the system.

9. Testimonials from team

Team Member 1 (Rahul):

“The project helped in improving analytical thinking and practical knowledge in web development, Spring Boot architecture, AI integration, and data visualization while working on a real-world enterprise application.” Team Member 2:

“Working on this project strengthened problem-solving abilities through tasks such as backend development, data processing, and implementing AI models for learning roadmaps.” Team Member 3:

“The development of the SkillSphere platform enhanced technical skills in React 19, Spring Boot, database management, and AI-based system design while providing experience in building real-world applications.”

10. Conclusion

The Enterprise Learning Platform — SkillSphere Nexus project successfully demonstrates the integration of web development and artificial intelligence to create an intelligent learning monitoring system. The platform allows users to track learning-related data such as course progress, quiz scores, and study streaks, while providing predictions and insights through an AI model. The system also includes an interactive dashboard, data visualization, and an AI chatbot to support users in understanding their learning patterns.

Through this project, practical knowledge of React 19, Spring Boot, MySQL database management, and machine learning/AI was applied to build a real-world application. Overall, the project highlights how technology can be used to promote better skill awareness and support healthier learning habits.

11. Acknowledgements

We express our sincere gratitude to Infosys Springboard for providing an excellent learning environment, industryoriented curriculum support, and access to valuable resources that helped in the successful development of the Enterprise Learning Platform — SkillSphere Nexus project.

We extend our heartfelt thanks to Shakthi Gopalakrishnan (Shakti Gopala Krishnan), our mentor, for continuous guidance, constructive feedback, and encouragement throughout the project development process. The insights and suggestions provided played a significant role in improving our understanding and refining the implementation of the system.